Product Scanning Application - Stakeholder Guide
Table of Contents
Executive Summary
At a Glance
Business Problem & Solution
The Problem
The Solution
Key Features
1. **QR Code Scanning with GS1 Support**
2. **Session-Based Scanning**
3. **Hospital & Batch Management**
4. **Expired Product Reporting**
5. **Hospital-to-Hospital (H2H) Stock Transfer**
6. **Dashboard & Reporting**
7. **Notifications & Alerts**
8. **Audit Trail & Compliance**
9. **Remarks & Comments**
10. **Master Data Management**
User Roles & Access
1. **PS Salesperson** (Field Scanner)
2. **PS Warehouse** (Dashboard Operator)
3. **Product Scanner** (Admin/Full Access)
4. **Sales** (Legacy Full Access)
5. **Superuser**
How It Works
Workflow 1: Stock Audit at Hospital (Normal Scan)
Workflow 2: Hospital-to-Hospital Transfer (H2H)
Workflow 3: Expired Product Detection
Use Cases
Use Case 1: Monthly Hospital Stock Audit
Use Case 2: Emergency Stock Transfer
Use Case 3: Expired Product Recall
Use Case 4: New Product Launch Distribution
Benefits & Value Proposition
For Sales Team
For Warehouse/Operations
For Management
For the Business
Technical Overview
Technology Stack
Data Model
Security Features
Integration Points
Getting Started
For Salespersons (Scanner Access)
For Warehouse Managers (Dashboard Access)
For Administrators (Full Access)
FAQs
General Questions
For Salespersons
For Warehouse Managers
Technical Questions
Support & Contact
Appendix A: GS1 Application Identifiers Reference
Appendix B: Session Status Lifecycle
Appendix C: User Permission Matrix
Document Version History

Product Scanning Application - Stakeholder Guide

Version 1.0
Date: May 2026
NexGen E-Meril Platform

Table of Contents

1. Executive Summary
2. Business Problem & Solution
3. Key Features
4. User Roles & Access
5. How It Works
6. Use Cases
7. Benefits & Value Proposition
8. Technical Overview
9. Getting Started
10. FAQs

Executive Summary

The **Product Scanning Application** is a comprehensive inventory tracking and management system designed for Meril Life Sciences to monitor medical device inventory across hospitals and distribution channels. The application enables real-time tracking of products through QR code scanning, providing complete visibility of stock location, expiry dates, and product movement between hospitals.

At a Glance

- **Purpose**: Track medical device inventory at hospitals using GS1-compliant QR code scanning
- **Users**: Sales personnel (field scanners) and warehouse managers (dashboard operators)
- **Primary Functions**: Stock audits, expired product detection, hospital-to-hospital transfers
- **Technology**: AI-powered GS1 barcode parser with OpenAI enhancement
- **Impact**: Eliminates manual tracking, reduces expired stock, improves inventory visibility

---

Business Problem & Solution

The Problem

Medical device companies face critical challenges in inventory management:

1. **Lack of Visibility**: No real-time knowledge of which products are

at which hospitals
2. **Expired Products**: Medical devices expire and need tracking to prevent usage beyond expiry dates
3. **Stock Movement**: Hospital-to-hospital transfers are difficult to track and verify
4. **Manual Processes**: Paper-based inventory tracking is error-prone and time-consuming
5. **Compliance Risk**: Regulatory requirements demand accurate product tracking and expiry management
6. **Lost Revenue**: Products lost in the field or expired at hospitals represent financial loss

The Solution

The Product Scanning Application provides:

- **Digital Scanning**: QR code-based product scanning using mobile devices or desktop cameras
- **Automated Tracking**: Automatic capture of product details (serial number, batch, expiry date, material code)
- **Smart Parsing**: AI-powered GS1 barcode parser that extracts all product information automatically
- **Real-Time Visibility**: Dashboard showing all scanned products, their locations, and status
- **Expiry Monitoring**: Automatic detection and reporting of expired or near-expiry products
- **Transfer Tracking**: Complete audit trail of stock movements between hospitals
- **Role-Based Access**: Separate interfaces for field personnel and warehouse managers

---

Key Features

1. QR Code Scanning with GS1 Support

- Scans GS1 DataMatrix and standard barcodes
- Automatically extracts:
  - Serial Number (AI 21)
  - GTIN/Material Code (AI 01)
  - Batch/Lot Number (AI 10)
  - Expiry Date (AI 17)
  - Production Date (AI 11)
  - Best Before Date (AI 15)
  - Packaging Date (AI 13)
  - Reference Numbers (AI 240/22)
- AI-powered parsing fills missing fields using OpenAI
- Fallback regex parsing for offline scenarios

2. Session-Based Scanning

- Create scanning sessions tied to specific hospitals and batches
- Two session types:
  - **Normal Scan**: Stock audit at a hospital
  - **H2H Transfer**: Hospital-to-hospital stock movement
- Scan multiple products without re-entering hospital/batch information

- Session states: Active → Completed → Closed
- Duplicate prevention (same serial within 30 seconds)

3. Hospital & Batch Management

- Master data for hospitals with SAP integration
- Hospital details: Name, SAP code, province, location, contact person, salesperson
- Batch grouping for organizing multiple scanning sessions
- GTIN-to-Material mapping for automatic material code lookup

4. Expired Product Reporting

- Automatic detection of expired products during session completion
- Consolidated expired product report with:
  - Days expired calculation
  - Hospital and session context
  - Acknowledgment workflow for warehouse team
- Email notifications to warehouse managers

5. Hospital-to-Hospital (H2H) Stock Transfer

- Dedicated transfer workflow separate from normal scanning
- Track stock movement from source to destination hospital
- Transfer ID format: `DDMMYY-Initials-FROM-TO-TOCODE`
- Complete transfer history and audit trail
- Confirmation attachment upload (delivery documents, receipts)

6. Dashboard & Reporting

For Warehouse Managers: - View all submitted scanning sessions - Session details with product lists - Expired product reports - Upload confirmation attachments - Terminate/force-complete sessions if needed - Export session data to Excel - H2H transfer tracking dashboard

For Salespersons: - Personal dashboard showing their sessions - Session status tracking - Ability to complete and submit sessions

7. Notifications & Alerts

- In-app notifications for warehouse managers
- Email alerts when sessions are submitted
- Email notifications when sessions are terminated
- Warehouse remark notifications
- Configurable recipient lists

8. Audit Trail & Compliance

- Complete audit log of all actions:
  - Session started/completed/terminated
  - Products scanned/removed
  - Attachments uploaded
  - Session exports
- Track who did what, when, and from where (IP address)
- User agent tracking for security
- Immutable history for regulatory compliance

9. Remarks & Comments

- Add remarks to individual scanned products
- Add remarks to entire scanning sessions
- Warehouse manager can add follow-up remarks
- Automated email notifications when warehouse adds remarks

10. Master Data Management

- Hospital master with full CRUD operations
- GTIN-Material mapping upload (Excel)
- Bulk import capabilities
- SAP integration ready

User Roles & Access

The application implements role-based access control (RBAC) with three primary roles:

1. PS Salesperson (Field Scanner)

Access: QR Scanner Interface Only

Responsibilities: - Create scanning sessions at hospitals - Scan product QR codes - Complete and submit sessions - View their own session history

Typical Users: Field sales representatives, territory managers

---

2. PS Warehouse (Dashboard Operator)

Access: Dashboard Interface Only

Responsibilities: - Review submitted scanning sessions - View all products scanned across sessions - Upload confirmation attachments to close sessions - Monitor expired product reports - Acknowledge expired products - Terminate sessions if needed - Export data for reporting

Typical Users: Warehouse managers, inventory managers, operations team

---

3. Product Scanner (Admin/Full Access)

Access: Both Scanner + Dashboard

Responsibilities: - All scanner functions - All dashboard functions - Manage master data (hospitals, GTIN mappings) - Force-complete sessions - Full administrative control

Typical Users: Product managers, system administrators, senior operations staff

---

4. Sales (Legacy Full Access)

Access: Both Scanner + Dashboard (backward compatibility)

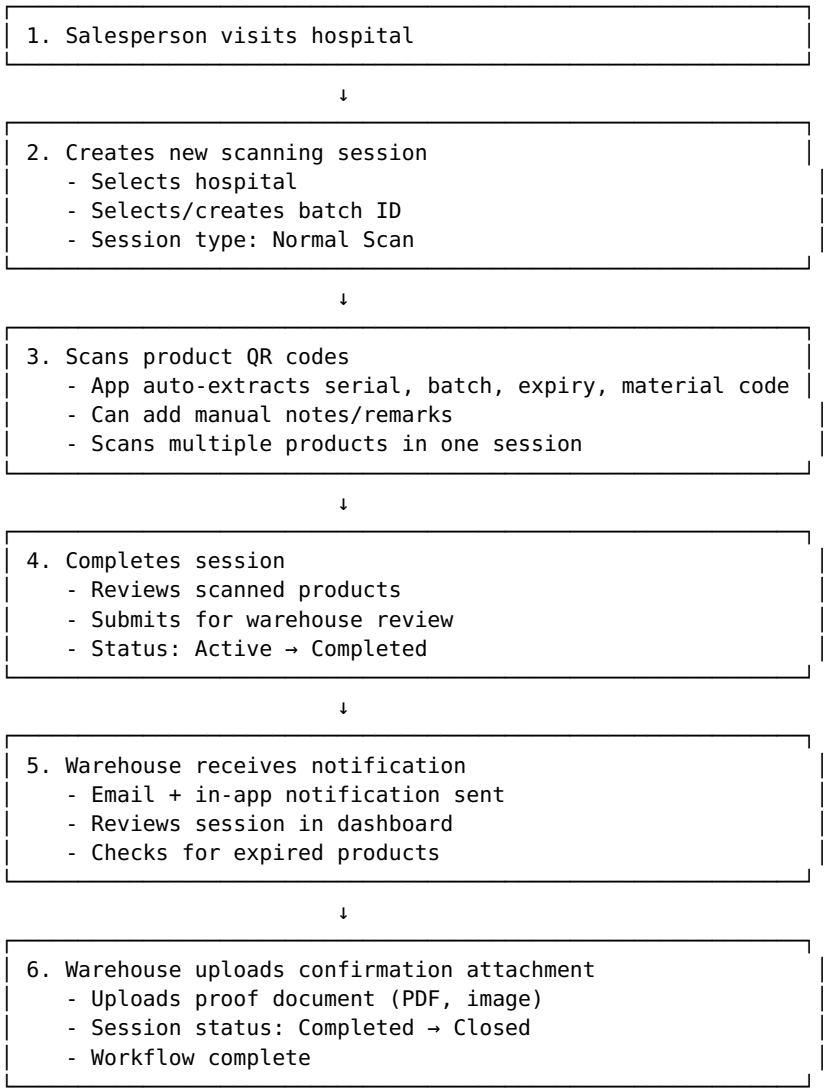
---

5. Superuser

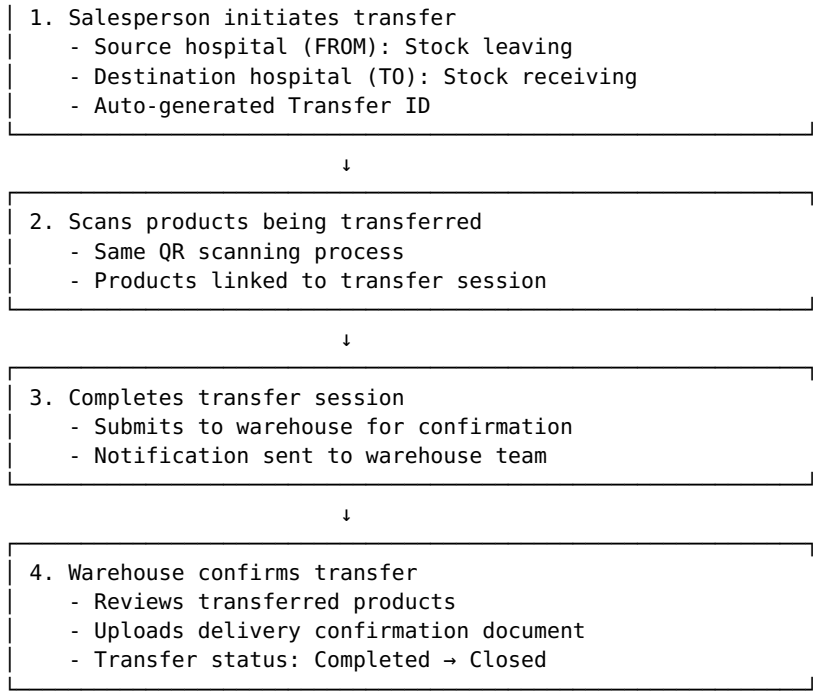
Access: Full system access including Django admin

How It Works

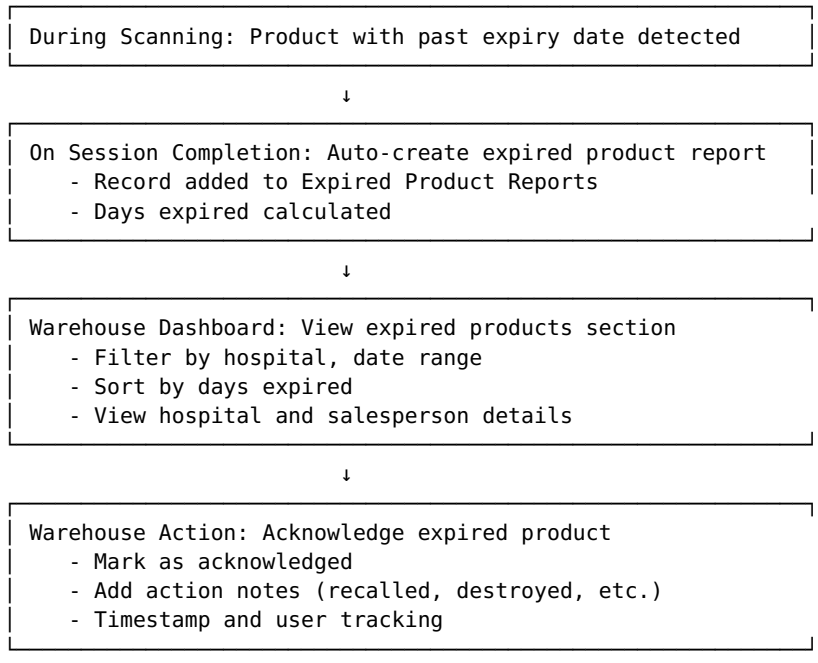
Workflow 1: Stock Audit at Hospital (Normal Scan)



Workflow 2: Hospital-to-Hospital Transfer (H2H)



Workflow 3: Expired Product Detection



Use Cases

Use Case 1: Monthly Hospital Stock Audit

Scenario: Sales team conducts monthly inventory audit at City Hospital

Steps: 1. Salesperson arrives at hospital with mobile device 2. Opens Product Scanner app, creates new session 3. Selects “City Hospital” and batch “MAY2026-AUDIT” 4. Walks through hospital storage, scanning all Meril products 5. App auto-captures: Serial numbers, expiry dates, batch numbers 6. Completes session with 147 products scanned 7. Warehouse receives notification, reviews inventory 8. Identifies 3 products expired, initiates recall 9. Uploads confirmation, closes session

Outcome: Complete inventory visibility, expired stock identified and removed

---

Use Case 2: Emergency Stock Transfer

Scenario: Hospital A has excess catheters, Hospital B needs them urgently

Steps: 1. Salesperson creates H2H transfer: Hospital A → Hospital B 2. Scans 25 catheters being transferred 3. Submits transfer session 4. Products physically moved to Hospital B 5. Warehouse confirms receipt with delivery document 6. Transfer marked as closed

Outcome: Stock movement tracked, both hospitals’ inventories updated, audit trail maintained

---

Use Case 3: Expired Product Recall

Scenario: Warehouse identifies 6-month-old scanning sessions with expired products

Steps: 1. Warehouse opens Expired Products Report dashboard 2. Filters by date range: Last 6 months 3. Exports report to Excel 4. Identifies 23 expired products across 5 hospitals 5. Coordinates with sales team for product recall 6. Marks products as acknowledged with recall notes

Outcome: Proactive expired product management, compliance maintained

---

Use Case 4: New Product Launch Distribution

Scenario: Meril launches new stent, distributed to 50 hospitals

Steps: 1. Sales team receives batch IDs for the launch: “STENT-2026-LAUNCH” 2. Each salesperson scans delivered products at their hospitals 3. All sessions linked to same batch ID 4. Warehouse dashboard shows real-time distribution status 5. Management sees: 50 hospitals, 2,300 stents, all sessions closed

Outcome: Launch tracking, distribution visibility, proof of delivery

---

Benefits & Value Proposition

For Sales Team

- □ **Faster scanning**: No manual data entry, automatic field extraction
- □ **Mobile-friendly**: Scan from anywhere using phone camera
- □ **Error reduction**: AI-powered parsing minimizes mistakes
- □ **Session history**: Track all past scans and sessions
- □ **Proof of work**: Digital record of hospital visits and stock checks

For Warehouse/Operations

- □ **Real-time visibility**: Know exactly what's where at any time
- □ **Expired product alerts**: Automatic detection and reporting
- □ **Audit trail**: Complete history of all stock movements
- □ **Data export**: Excel exports for analysis and reporting
- □ **Workflow automation**: Email notifications, automated status updates
- □ **Transfer tracking**: H2H stock movement visibility

For Management

- □ **Inventory insights**: Data-driven decisions on stock allocation
- □ **Compliance**: Regulatory-compliant product tracking
- □ **Cost reduction**: Minimize expired stock losses
- □ **Performance metrics**: Track scanning activity by salesperson, hospital, region
- □ **Scalability**: System handles growing product and hospital volume

For the Business

- □ **Reduced Losses**: Less expired stock = less write-offs
- □ **Better Forecasting**: Historical data enables demand prediction
- ⚡ **Faster Recalls**: Quick identification of affected products
- □ **Risk Mitigation**: Compliance with medical device regulations
- □ **Customer Satisfaction**: Hospitals trust accurate inventory management

---

Technical Overview

Technology Stack

Backend: - Django 4.x (Python web framework) - PostgreSQL database - Django REST Framework for APIs

Frontend: - HTML5, CSS3, JavaScript - Bootstrap for responsive UI - AJAX for real-time interactions

AI/ML: - OpenAI GPT integration for GS1 barcode enhancement - Custom GS1 Application Identifier parser - Regex fallback for offline scenarios

Infrastructure: - Docker containerization - Linux server deployment - Email service integration

Data Model

Core Entities: - `Hospital`: Hospital master data with SAP codes - `Batch`: Grouping mechanism for sessions - `ScanningSession`: A scanning event at a hospital - `ScannedProduct`: Individual product scan record - `TransferSession`: H2H stock transfer - `ExpiredProductReport`: Auto-generated expired product records - `GTINMaterialMapping`: GTIN to SAP material code lookup - `AuditLog`: Complete audit trail

Security Features

- Role-based access control (RBAC)
- Django authentication & authorization
- IP address tracking
- User agent logging
- Session security with unique keys
- XSS and CSRF protection

Integration Points

- **SAP**: Hospital SAP codes, Material codes ready for integration
- **Email**: SMTP integration for notifications
- **Export**: Excel/CSV export capabilities
- **OpenAI**: Optional AI-powered parsing enhancement

Getting Started

For Salespersons (Scanner Access)

Step 1: Access the Application - Navigate to: `https://nexgen.e-meril.in/product_scanning/` - Login with your credentials - You'll see the QR Scanning Form

Step 2: Create a Session 1. Select Hospital from dropdown 2. Enter or select Batch ID (e.g., `MAY2026-AUDIT-01`) 3. Session Type: Select "Normal Scan" (or "Transfer" for H2H) 4. For transfers, also select "To Hospital" 5. Click "Start Scanning Session"

Step 3: Scan Products 1. Click "Scan QR Code" to activate camera 2. Point camera at product QR code/barcode 3. App auto-fills: Serial, Batch, Expiry, Material Code 4. Verify information, add notes if needed 5. Click "Add Product" 6. Repeat for all products

Step 4: Complete Session 1. Review all scanned products in the list 2. Click "Complete Session" 3. Confirm submission 4. You're done! Warehouse team will be notified

For Warehouse Managers (Dashboard Access)

Step 1: Access Dashboard - Navigate to: `https://nexgen.e-meril.in/product_scanning/dashboard/` - Login with your credentials

Step 2: Review Sessions 1. View “Pending Sessions” (status: Completed) 2. Click on a session to view details 3. Review all scanned products 4. Check for expired products (highlighted in red)

Step 3: Close Sessions 1. Upload confirmation attachment (PDF/image) 2. Add notes if needed 3. Click “Upload & Close Session” 4. Session status changes to “Closed”

Step 4: Manage Expired Products 1. Navigate to “Expired Products Report” 2. Review list of expired items 3. Click “Acknowledge” on each item 4. Add action notes (recalled, disposed, etc.)

Step 5: Track H2H Transfers 1. Navigate to “Transfer Dashboard” 2. View all hospital-to-hospital transfers 3. Review transfer details and products 4. Upload delivery confirmation 5. Mark transfer as closed

For Administrators (Full Access)

Additional Capabilities: - Manage Hospital Master: `/product_scanning/dashboard/hospitals/` - Upload GTIN Mappings: `/product_scanning/dashboard/gtin-material-master/` - Force-complete sessions - Terminate sessions - View complete audit logs - Export data for analysis

---

FAQs

General Questions

Q: What types of barcodes does the system support?
A: The system supports GS1 DataMatrix, GS1-128, QR codes, and standard barcodes. It can parse GS1 Application Identifiers (AIs) automatically.

Q: Can I scan offline?
A: The app requires internet connection for full functionality. However, the GS1 parser has regex fallback that works without OpenAI.

Q: What if the QR code doesn’t scan properly?
A: You can manually enter the data. The app also shows the raw QR data for verification.

Q: How many products can I scan in one session?
A: There’s no hard limit. Sessions with hundreds of products have been tested successfully.

For Salespersons

Q: Can I edit a scanned product after adding it?
A: Yes, you can delete and re-scan, or add remarks. Once submitted, warehouse team can also add remarks.

Q: What happens if I accidentally close the browser during scanning?
A: The session remains active. Use the session key to resume or start a new session.

Q: Can I scan the same serial number twice?
A: The system prevents duplicate scans within 30 seconds. Legitimate duplicates (same SN on different dates) are allowed.

Q: Do I need to upload attachments?
A: No, salespersons submit sessions. Warehouse team uploads confirmation attachments to close them.

For Warehouse Managers

Q: How do I know when a session is submitted?
A: You receive both an email notification and an in-app notification.

Q: Can I reject a session?
A: Yes, you can add remarks or terminate a session if needed. The salesperson will be notified.

Q: What file types can I upload as attachments?
A: PDF, images (JPG, PNG), and common document formats. Files are stored securely in the database.

Q: How do I export data for reporting?
A: Each session detail page has an "Export to Excel" button. You can also export expired product reports.

Technical Questions

Q: Is the data backed up?
A: Yes, the PostgreSQL database has regular automated backups.

Q: What about data privacy and security?
A: The system follows industry standards: encrypted connections (HTTPS), role-based access, audit trails, and secure authentication.

Q: Can the system integrate with our ERP/SAP?
A: The data model includes SAP codes for hospitals and materials. API integration can be developed.

Q: What happens to data of closed sessions?
A: Data is retained indefinitely for compliance and historical reporting. Closed sessions cannot be modified.

Support & Contact

For technical support, feature requests, or questions:

Development Team: NexGen E-Meril Platform
Email: support@e-meril.in
Documentation: `/docs/` folder in the project repository

Appendix A: GS1 Application Identifiers Reference

The system automatically parses these GS1 AIs from barcodes:

AI	Description	Example
01	GTIN (Product Code)	00012345678905
10	Batch/Lot Number	BATCH2026A
11	Production Date (YYMMDD)	260507 (7 May 26)
13	Packaging Date (YYMMDD)	260430
15	Best Before Date (YYMMDD)	280507
17	Expiry Date (YYMMDD)	270507
21	Serial Number	SN123456789
22	Secondary Data	REF-12345
240	Additional Product Info	Additional ID

Appendix B: Session Status Lifecycle

```
ACTIVE
  ↓
  User clicks "Complete Session"
  ↓
COMPLETED (Pending warehouse review)
  ↓
  Warehouse uploads attachment
  ↓
CLOSED (Workflow complete)

Alternative paths:
ACTIVE → CANCELLED (User cancels before completion)
ACTIVE/COMPLETED → TERMINATED (Warehouse terminates session)
```

Appendix C: User Permission Matrix

Function	PS Salesperson	PS Warehouse	Product Scanner	Superuser
Scan QR Codes				
Create Sessions				
Complete Sessions				
View Dashboard				
Upload Attachments				
Terminate Sessions				
Manage Hospitals				
Upload GTIN Mappings				

View Expired Reports
Force-Complete Sessions
View Audit Logs
Export Data

Document Version History

Version	Date	Changes	Author
1.0	May 2026	Initial stakeholder guide	NexGen

End of Document

This guide is maintained as part of the NexGen E-Meril platform documentation.